

PRATAP KUMAR MISHRA

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience in data analysis, machine learning, and visualization gained through three internships and applied analytics projects. Proficient in Python (pandas, NumPy, Matplotlib, Seaborn), SQL, and Excel, with experience building predictive models and translating data into actionable insights. Seeking a Data Analyst role.

EDUCATION

B.Tech in Computer Science and Engineering Expected Graduation: 2027

Nalanda Institute of Technology, Bhubaneswar | CGPA: 7.96 / 10

TECHNICAL SKILLS

- **Programming & Query Languages:** Python, SQL, C, HTML/CSS
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, MS Excel
- **Machine Learning:** Linear Regression, Logistic Regression, Random Forest, Gradient Boosting, scikit-learn
- **Model Evaluation & Concepts:** MAE, RMSE, R^2 , Precision, Recall, F1-Score, ROC-AUC, Data Cleaning, EDA, Feature Engineering, Statistical Analysis
- **Tools & Platforms:** Jupyter Notebook, VS Code, Git/GitHub, ESP32 (IoT)

INTERNSHIP EXPERIENCE

Data Analysis & AI/ML Intern | Xylofy AI Jun 2026 – Jul 2026

- Built and evaluated 3 end-to-end data analytics projects covering predictive modeling, classification, and algorithmic trading using Python, pandas, NumPy, and scikit-learn (detailed under Projects).
- Performed data cleaning, missing value imputation, feature encoding, and exploratory data analysis (EDA) across multiple real-world datasets.
- Created visual reports including correlation heatmaps, ROC curves, confusion matrices, and trend charts using Matplotlib and Seaborn to communicate findings.

IoT & Data Analysis Intern | NIT Rourkela Jun 2025 – Jul 2025

- Designed and implemented an IoT-based automatic curtain system using an ESP32 microcontroller, LDR and temperature sensors, and a servo motor for automated, sensor-driven control.
- Collected and analyzed sensor data to calibrate automation thresholds, improving system responsiveness and reliability.

PROJECTS

House Price Prediction | Python, pandas, scikit-learn, Matplotlib, Seaborn

- Built a regression pipeline using Linear Regression and Random Forest ($n_estimators=200$) to predict house prices, with preprocessing including missing value imputation, duplicate removal, binary and one-hot encoding, and an 80/20 train-test split. Evaluated and compared performance using MAE, RMSE, and R^2 ; visualized price distribution, feature correlation heatmap, actual vs. predicted prices, and feature importance.

Employee Attrition Prediction | Python, scikit-learn, Seaborn, StandardScaler

- Developed a classification system using Logistic Regression, Random Forest, and Gradient Boosting to predict employee attrition, applying feature scaling and class balancing ($class_weight='balanced'$) to handle imbalanced data.
- Evaluated models using Precision, Recall, F1-score, and ROC-AUC, selecting the best-performing model by ROC-AUC; identified salary level, work-life balance, tenure, department, and job role as the strongest attrition drivers through EDA and feature importance analysis.
- Visualized department- and role-wise attrition trends, income distributions, confusion matrices, and ROC curve comparisons across models.

Algorithmic Stock Trading Strategy | Python, yfinance, pandas, Matplotlib

- Implemented a rule-based trading strategy combining Moving Average crossover (MA20/MA50), 14-period RSI momentum filtering, and Bollinger Bands volatility analysis on one year of historical daily stock data; backtested against a buy-and-hold benchmark using a simulated \$10,000 portfolio to calculate strategy return, benchmark return, and alpha. Visualized price trends, RSI levels, trading volume, and portfolio performance versus the benchmark.

CERTIFICATIONS & ACTIVITIES

- Python for Beginners — freeCodeCamp | Participant, AI/ML Event — Nalanda Institute of Technology

ACHIEVEMENTS

- Completed three end-to-end machine learning and data analytics projects.
- Published projects on GitHub using Git version control and documentation.